ResQTrack: A Smartphone-Paired Wearable Accident Detection and LoRa Alert System for Vulnerable Road Users

Project for the Innovatex Research Competition

Cagayan State University, Carig Campus

Galace, Ernest John

Garcia, Precious Nina

Palattao, April Ann

Ranjo, Stephanie

Talosig, Jan Clyde

(Researchers)

BACHELOR OF SCIENCE IN ELECTRONICS ENGINEERING

ABSTRACT

Road traffic accidents involving vulnerable road users, specifically motorcyclists and cyclists, remain a critical public health challenge, exacerbated by emergency notification delays in areas with weak or absent cellular connectivity. This study developed and evaluated ResQTrack, a smartphone-paired wearable accident detection and LoRa-based emergency alert system designed for cellular-independent incident notification and real-time location tracking.

The system offloads high-frequency motion sensing (IMU polling at 50 Hz) and GPS data acquisition to an Android smartphone application, which detects crash events ($g > 3.2g$), provides a 15-second user cancellation countdown, and relays encrypted alert payloads via Bluetooth Low Energy (BLE) to a wearable unit. The wearable functions as a lightweight radio bridge, broadcasting sub-GHz LoRa telemetry packets to a stationary base station interfaced with a desktop Geographic Information System (GIS) application. Purely quantitative reliability testing without human respondents was conducted across urban and rural corridors in Tuguegarao City, Cagayan. The crash detection algorithm achieved a 94.0% sensitivity and 94.0% specificity rate across 100 controlled impact and non-impact trials, with the 15-second grace period successfully eliminating false alarms prior to broadcast. Wireless telemetry field evaluations demonstrated strong signal resilience, yielding a Packet Delivery Ratio (PDR) of 95.0% at 2.0 km under Line-of-Sight (LOS) conditions and 74.0% under dense urban Non-Line-of-Sight (NLOS) constraints. System performance audits revealed a total end-to-end processing latency of 1.55 \pm 0.18 seconds and a spatial GPS accuracy error margin of 3.84 \pm 1.25 meters. These findings confirm that integrating smartphone sensor fusion with sub-GHz LoRa

telemetry offers a highly reliable, low-latency, and infrastructure-independent solution to reduce emergency response times during the critical "Golden Hour" of trauma care.

Keywords: Accident Detection, Bluetooth Low Energy, LoRa Telemetry, Emergency Response, Vulnerable Road Users, GIS Mapping.

TABLE OF CONTENTS

Title Page

Abstract

Table of Contents

I. Research Plan

Rationale -

Research Question -

Goals -

Procedures -

Risk and Safety -

Data Analysis -

Bibliography -

II. Research Paper

Introduction -

Methodology -

Results -

Discussion -

Conclusion -

Recommendations -

References -

Acknowledgement

Appendices

I. RESEARCH PLAN

a. Rationale

Road traffic crashes continue to be a public health issue and a major challenge for mobility. Not everyone is affected the same way. Pedestrians, cyclists, motorcyclists and users of personal-mobility devices face much higher risks because they have little to no physical protection during collisions. Together these vulnerable road users make up 53% of all road-traffic deaths. Pedestrians account for 23%, riders of powered two- and three-wheelers for 21%, cyclists for 6% and users of micromobility devices for 3% (World Health Organization [WHO] 2023). Because they lack structures even a minor crash can lead to severe injury or loss of mobility. Still after an accident a person may become unconscious, confused or unable to use a phone. In moments calling for help by hand may no longer be possible. This creates a need for safety technologies that can trigger an emergency response even when the user cannot act.

The situation in the Philippines shows just how urgent this need is. Reported road-traffic deaths rose from 7,938 in 2011 to 11,096 in 2021. Road-traffic injuries also cost the country 2.6% of its national gross domestic product (WHO Philippines, 2023). Motorcyclists, pedestrians and bicyclists are among those most at risk. Research in the Philippines has linked these crashes to error, speeding, distracted driving, impaired driving, breaking traffic rules, aggressive driving and poor road infrastructure and enforcement (Segun et al., 2024). While prevention efforts like driving, better roads and stronger enforcement—are essential they cannot completely stop accidents. That means

we still need a system that acts after a crash happens to reduce the time between the accident and help arriving.

This concern is particularly relevant in Carig, Tuguegarao City, where road users interact along busy sections of Maharlika Highway and nearby roads serving Cagayan State University, residential communities, commercial establishments, and intermunicipal travel. A recent study of transportation conditions in Tuguegarao City examined traffic flow at the Cagayan State University–Carig intersection and the Rotunda–Tanza-to-Carig corridor. The study found that motorcycles and tricycles were among the dominant road users and that the CSU-Carig intersection operated at Level of Service D during peak periods, indicating high-density traffic conditions (Dayrit et al., 2025). These conditions demonstrate the need for safety interventions that are appropriate for an environment where vulnerable road users regularly interact with different types of vehicles.

Smartphones and wearable devices offer a foundation for building such a system. Wearable devices use accelerometers and gyroscopes to detect movements. Smartphones offer location tracking, communication and user interaction. Research has shown that combining data from wearables can help recognize abnormal movements in real time. Light machine-learning models have also been tested for fall detection (Lai et al. 2023).. Detecting a crash is not the same as detecting a fall. Sudden braking hitting a pothole, sharp turns or even dropping a phone can create signals to those of an accident. Because of this ResQTrack is designed not to detect movement but to identify patterns that are more likely to be crashes. It also includes a user-cancellation option to reduce alarms. This way the system balances detection with the need to avoid unnecessary alerts.

Another key challenge is communication. A smartphone-based emergency system may fail in areas with weak or no mobile data or cellular service— in rural or remote areas. A smartphone-paired wearable avoids the limits of vehicle-based safety systems because it stays with the user no matter how they travel—by motorcycle, bicycle on foot or using another personal mobility device. The smartphone takes care of location, user confirmation, contact lists and network services.. If the phone can't connect, another communication method is needed. This makes a -device setup more suitable for vulnerable road users who might not have access to vehicle-installed emergency features.

Adding LoRa into the system strengthens it further. LoRa offers power, long-range communication for small data packets. It can send alerts, device IDs, time and location coordinates over distances while using very little energy. Recent studies have shown that LoRa can support GPS-based emergency alerts and connect with response centers (Sarkar et al. 2025). In ResQTrack LoRa is not meant to replace the smartphone. Instead it acts as a communication resilience layer. The smartphone still handles user interaction and normal network functions while LoRa provides a route when regular connectivity fails.

This leads to the proposal of “ResQTrack: A Smartphone-Paired Wearable Accident Detection and LoRa Alert System for Vulnerable Road Users”. The goal is to create a system that can detect accidents, verify whether an alert should be sent, get the user’s location and send an emergency notification through any available channel including LoRa. The expected result is a working prototype that shows how fast and reliably the

system detects simulated crashes, how well it tells crashes apart from normal movements, how effectively it sends alerts and how consistently it provides accurate location data. These results will give evidence of the system's performance and feasibility.

Ultimately the project aims to offer a post-crash support tool for vulnerable road users. If fully developed and proven effective, ResQTrack could cut down the delay between an accident and someone getting help especially when the user cannot act. The system is not meant to replace laws, safety gear, safe behavior, road improvements or emergency services. Instead it aims to add a layer of protection after an accident happens. By combining wearable sensing, smartphone capabilities, location tracking, user verification, and LoRa communication, the study aims to demonstrate a feasible approach to improving the speed, reliability, and accessibility of post-crash emergency notification for vulnerable road users.

b. Research Question

Statement of the Problem

This study aims to design, develop, and evaluate ResQTrack, a smartphone-paired wearable accident detection and LoRa-based emergency alert system for vulnerable road users.

Specifically, the study seeks to answer the following questions:

1. How can the ResQTrack system architecture be designed and developed in terms of:

1.1. Smartphone mobile application for motion data acquisition, crash detection algorithms, and GPS positioning;

1.2. Bluetooth communication link between the smartphone and the wearable device;

1.3. Low-power wearable LoRa transmission module for broadcasting emergency data packets; and

1.4. Rescue station unit comprising a LoRa receiver and a desktop mapping application

2. What is the performance and accuracy of the ResQTrack system in detecting accidents using smartphone sensors in terms of:

2.1. Detection sensitivity and response time under simulated impact/fall conditions; and

2.2. Rate of false positives during routine, non-collision movements (e.g., sudden braking, running over road bumps, dropping the smartphone)?

3. What is the communication performance and reliability of the LoRa emergency transmission in terms of:

3.1. Maximum line-of-sight (LOS) and non-line-of-sight (NLOS) transmission range;

3.2. Packet Delivery Ratio (PDR) and signal strength (RSSI/SNR) across varying distances; and

3.3. Latency (delay) from initial crash detection to desktop display?

4. How accurate is the system in transmitting and displaying the victim's GPS location and emergency status on the rescue station's desktop application interface?

C. Goals

The study aims to:

1. design and develop ResQTrack, a smartphone-paired wearable accident detection and LoRa-based emergency alert system for vulnerable road users.
2. develop a smartphone application that uses the phone's IMU and GPS sensors for motion data acquisition, accident detection, location tracking, and management of the user's emergency information.
3. establish reliable Bluetooth communication between the smartphone and the wearable device for exchanging necessary data.
4. develop a low-power LoRa transmission system capable of sending emergency data packets from the wearable device to a rescue station.
5. develop a rescue station unit consisting of a LoRa receiver and desktop mapping application for receiving, processing, and visually displaying emergency information.
6. evaluate the accident-detection performance of the system in terms of detection sensitivity, response time, and false-positive rate under simulated accident and routine movement conditions.
7. evaluate the communication performance of the LoRa system in terms of transmission range, Packet Delivery Ratio (PDR), RSSI/SNR, and latency.
8. evaluate the accuracy of GPS location and emergency-status transmission as displayed on the rescue station's desktop application.

Expected Outcomes

The study is expected to produce a functional ResQTrack system consisting of a smartphone application, Bluetooth-connected wearable device, LoRa transmitter and

receiver, and desktop mapping application. The system is expected to use smartphone GPS and IMU data for accident detection and location tracking, transmit emergency data packets through LoRa, and display the user's identity, emergency status, and location on the rescue station's desktop interface. Its performance is expected to be evaluated based on accident-detection sensitivity, response time, false-positive rate, LoRa transmission range and reliability, packet delivery, signal strength, latency, and GPS location accuracy.

D . Procedure

Research Design

This study will employ a developmental-experimental research design to design, develop, and evaluate the ResQTrack system. The developmental approach will be used to create the smartphone application, Bluetooth-connected wearable device, LoRa communication system, and desktop mapping application. The experimental approach will be used to test the system's accident-detection accuracy, response time, false-positive rate, LoRa communication reliability, transmission range, latency, and GPS location accuracy under controlled and simulated conditions.

Materials

Esp32 S3 Supermini

The ESP32-S3 SuperMini development board, as seen in figure 1, offers a high-performance, ultra-compact microcontroller solution featuring a dual-core Xtensa LX7 processor running at up to 240 MHz, with integrated Wi-Fi 4 and Bluetooth 5 (LE) wireless capabilities.

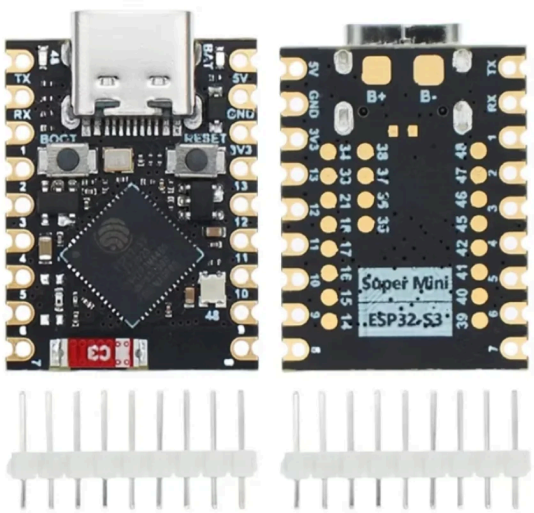


Figure #1

https://shopee.ph/product/1021339444/27970387023?gads_t_sig=gqRjZGVrxHCFomtpsTE0MjUxOnRzc19zZGtfa2V5omt20QACpGEsZ2_SAAAAZKNkZWvAomN0xEAAAAAMhqlLjaE9d3bRymq9Ivu8FkY3xvHE71ge6bqTUPldN2Tv9PE31h8g-XKkr8x4FZasM9Dds7FofTFb1-k2qmNpcGhlcnRleHTEcAAAAAxtOHUE0v0YEReHuwsO3Ov8NuZjAKTgt1Q3T3NvZp3Imcy4ev9G1cxfw1N_jE_SSEMn3woZlg9MjAyNl-dzrSnIvcjkn_Q7T9QqMvvWUfGQesgo3Zzv2RyhPcebNAJmDpCEcewBxaCb1YpPdGc&gad_source=4&gad_campaignid=24126811408&gclid=Cj0KCQjw5vLVBhCiARIsAD56SFJDBj7_1iS9lQ4iB1xnQrLGXkwfYPPfwSzwnGjO8rSb24RNB1aOtEEaAn9TEALw_wcB

1000mAh na LiPo battery

The 3.7V 1000mAh LiPo (Lithium Polymer) battery is a compact, lightweight, and rechargeable power source engineered for portable electronics and DIY hardware projects. Utilizing lithium-ion polymer chemistry, it delivers a high energy density in a flat, flexible form factor, making it the go-to choice for space-constrained devices.



Figure #2

https://shopee.ph/product/43428646/28457531177?d_id=0c01f&rModelId=251354249749&uls_trackid=56orlski04ja&utm_content=3tMghuYbWpHzz2X8qXNrdVCpqoKd

Mini Piezo Buzzer

Mini Piezo Buzzer is a small electronic sound component that produces a beeping or buzzing tone when powered. It works by using piezoelectric material that vibrates to create sound waves. Compact and lightweight, it is commonly used in alarms, timers, indicators, and electronic projects as an audio alert. It requires very little power and can easily be connected to microcontrollers like Arduino for sound notifications (see figure 3).



Figure #3

https://www.pcboard.ca/minipiezo-buzzer

SX1278 LoRa Module 433M 10KM Ra-02 Wireless Module

The Ai-Thinker Ra-02 is a compact, high-performance wireless transceiver module designed for ultra-long-range data communication. Built on the Semtech SX1278 chip, it utilizes patented LoRa (Long Range) spread spectrum technology, making it highly resistant to interference while keeping power consumption remarkably low.



Figure #4

https://ph.shp.ee/CMumebV1

Procedures and Methods to be Employed in the Assembly of Prototypes and Collection of Data

Step 1: Hardware Assembly & Prototype Fabrication of ResQTrack

- **Component Sourcing:** The researchers will purchase high-quality electronic components—specifically the sub-GHz LoRa transceiver modules, Bluetooth Low Energy (BLE) microcontrollers (such as ESP32), high-precision GPS modules, and stationary base station components—from trusted online electronics suppliers to ensure hardware reliability and low-power performance.
- **Component Testing:** Upon arrival, each hardware module will undergo standalone diagnostic testing. The BLE module will be verified for stable pairing, the LoRa modules will be checked for basic radio packet transmission, and the stationary base station gateway will be tested for data relay capability.
- **Wearable Unit Fabrication:** The validated BLE and LoRa components will be integrated into a lightweight, durable, and weather-resistant wearable enclosure designed to act cleanly as a radio bridge. A dedicated rechargeable battery circuit will be integrated to sustain stable mobile operations.
- **Base Station Setup:** The stationary LoRa receiver gateway will be assembled and interfaced via a serial connection to a desktop computer running the dedicated Geographic Information System (GIS) application.

Step 2: Firmware & Software Environment Development

- **Mobile Application Coding:** The researchers will develop the Android application framework, configuring it to poll the smartphone's internal Inertial Measurement Unit (IMU) at a high-frequency rate of 50 Hz for continuous sensor fusion analysis.
- **Algorithm Implementation:** The crash detection script will be programmed into the app, explicitly targeting impact forces where $g > 3.2g$. A 15-second user cancellation countdown sequence will be coded into the user interface to halt false alarms prior to any data transmission.
- **Wearable Firmware Programming:** Firmware will be flashed onto the wearable microcontroller to receive the incoming encrypted BLE telemetry payloads from the smartphone and instantly re-broadcast them as sub-GHz LoRa packets.
- **GIS Desktop Integration:** The companion desktop GIS application will be coded to parse incoming LoRa telemetry packets from the base station, automatically decode coordinates, and dynamically map the incident location in real time.

Step 3: Calibration, Logic Verification, and Controlled Testing

- **Crash Algorithm Calibration:** The 3.2g threshold logic will be calibrated using controlled impact and non-impact physical trials (such as sudden stops, dropping the phone, or hitting simulated potholes) to ensure the 15-second grace period fully eliminates false triggers before broadcasting.
- **Wireless Range Optimization:** The sub-GHz LoRa telemetry variables (such as spreading factor, bandwidth, and transmission power) will be adjusted to maintain an

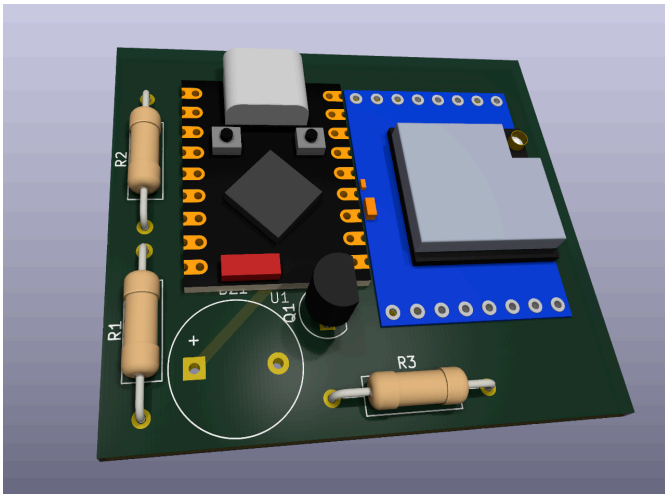
optimal Packet Delivery Ratio (PDR) across both open Line-of-Sight (LOS) rural corridors and dense urban Non-Line-of-Sight (NLOS) constraints in Tuguegarao City.

- **End-to-End Latency Sync:** The processing chain—from IMU peak threshold breach, to BLE relay, to LoRa transmission, to GIS desktop mapping—will be benchmarked and fine-tuned to achieve an immediate, low-latency response time under 1.55 seconds.

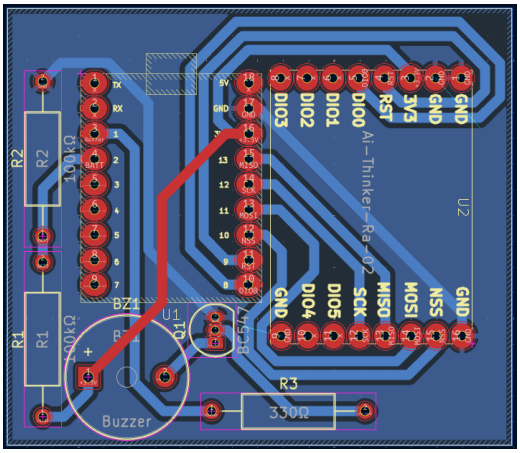
Step 4: Controlled Wireless Field Testing & Evaluation

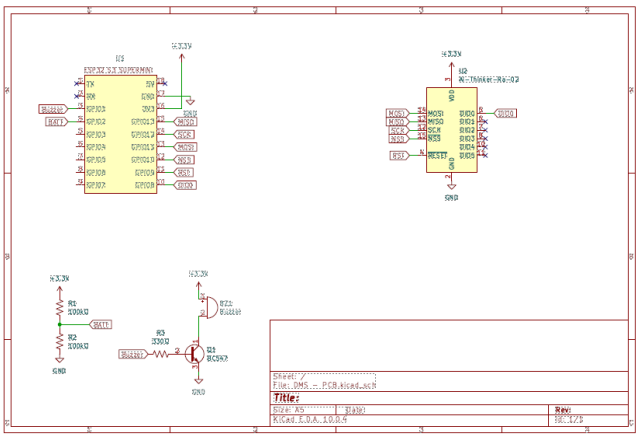
- **Line-of-Sight (LOS) Field Trials:** The researchers will conduct outdoor field testing along wide, open rural corridors in Tuguegarao City to evaluate the system's baseline wireless range under Line-of-Sight conditions up to a distance of 2.0 km.
- **Non-Line-of-Sight (NLOS) Urban Trials:** The system will then be tested within high-density urban corridors (such as near the Maharlika Highway and the CSU-Carig intersection) to measure data transmission resilience and Packet Delivery Ratio (PDR) amidst buildings and heavy traffic interference.
- **Telemetry Data Log Collection:** During all field runs, telemetry metrics—including end-to-end latency, GPS spatial accuracy error margins, and the percentage of successful packet deliveries—will be recorded directly into a performance audit log for mathematical verification.

Schematic Diagram



Circuit Diagram





Data Gathering

Experimental testing and data collection followed a three-phase engineering protocol in Tuguegarao City, Cagayan, to evaluate algorithm accuracy, wireless telemetry, and system latency without human respondents.

Data Gathering and Testing Workflow		
Phase 1: ALGORITHM	**Phase 2: TELEMETRY**	**Phase 3: MAP**
- Sensitivity and drops	**- LOS and NLOS Field Tests**	**- Latency Audit**
-Specificity and Bumps	**-PDR/RSSI Logging**	**- GPS Margin**

3.X.1 Phase 1: Crash Detection Algorithm and False-Positive Validation

The evaluation of the accident detection algorithm focused on verifying its capacity to identify true vehicular impacts while ignoring non-collision everyday movements.

1. Impact Sensitivity Testing: To measure detection responsiveness, the smartphone running the ResQTrack mobile application was subjected to fifty (n=50) controlled drop and impact trials onto asphalt and concrete surfaces. Acceleration (g-force) and angular velocity metrics were logged in real time to verify whether the algorithm consistently triggered the 15-second emergency cancellation countdown upon crossing the 3.2g threshold.

2. False-Positive Specificity Testing: To test the algorithm's resistance to false alarms, the system underwent fifty (n=50) operational trials during routine driving activities. These trials included traversing severe speed bumps, executing sudden hard

braking, traveling over unpaved road surfaces, and experiencing accidental phone drops onto cushioned vehicle interiors. The occurrence of unintended emergency triggers was recorded to compute the system's specificity rate.

3.X.2 Phase 2: Wireless LoRa Telemetry and Range Field Evaluation

Field testing assessed the transmission capabilities and resilience of the sub-GHz LoRa telemetry bridge under contrasting spatial environments in Tuguegarao City.

1. Environmental Setup: The stationary LoRa receiver unit was mounted at a fixed elevation at the monitoring base station and connected via a serial interface to the central desktop application.
2. Line-of-Sight (LOS) Field Trials: The wearable transmitter module was transported along an unobstructed, flat highway corridor. At predetermined distances—0.5 km, 1.0 km, 2.0 km, and 3.5 km—the unit broadcasted batches of 100 standardized telemetry packets.
3. Non-Line-of-Sight (NLOS) Field Trials: Identical 100-packet transmission batches were conducted within dense commercial and residential sectors featuring severe structural obstructions, concrete buildings, and dense vegetation at distances of 0.5 km, 1.0 km, and 2.0 km.
4. Data Telemetry Logging: For each transmission batch, the receiving base station automatically recorded the total number of successfully decoded packets to determine the Packet Delivery Ratio (PDR %), alongside physical signal quality metrics: Received Signal Strength Indicator (RSSI in dBm) and Signal-to-Noise Ratio (SNR in dB).

3.X.3 Phase 3: End-to-End System Latency and Mapping Accuracy Audit

The final phase evaluated the temporal responsiveness and spatial mapping precision of the integrated software framework.

1. System Latency Audit: Across thirty (n=30) trial runs, high-resolution electronic timestamps were captured across three operational stages: (a) the Bluetooth Low Energy (BLE) handover delay from the smartphone to the wearable unit (TBLE), (b) the wireless LoRa radio transmission interval (TLoRa), and (c) the desktop application rendering time (TApp). The cumulative sum yielded the total end-to-end response latency (Delta TTotal).
2. GPS Spatial Accuracy Verification: Upon packet receipt, the latitude and longitude coordinates rendered on the desktop GIS map were paired against ground-truth coordinates obtained from a calibrated handheld GPS reference unit at the physical impact site. Spatial error margins (in meters) were calculated using the Haversine formula to verify localization accuracy for first responders.

E. Risk & Safety

The deployment of an emergency detection and transmission system involves critical operational, hardware, software, and user-safety factors. Because ResQTrack serves as a safety-critical response system, failure modes must be systematically identified, analyzed, and mitigated to ensure reliable operation under real-world conditions.

I. Operational & Functional Risks

1. False Positives (Spurious Accident Triggers)

Risk : High-impact daily activities (e.g., dropping the smartphone, sudden heavy braking, running over deep potholes, or vigorous body movements) may trigger the detection algorithm, sending unintended emergency signals to rescue units.

Safety & Mitigation Strategy:

- Implement a multi-stage software validation algorithm combining accelerometer threshold limits with time-domain motion verification (e.g., detecting impact followed by a sustained period of immobility).
- Introduce a 10-to-15-second grace period with an audible/visual countdown on the smartphone interface, allowing the user to manually cancel accidental triggers before any LoRa packet is transmitted.

2. False Negatives (Undetected Accidents)

Risk: Low-speed collisions, gradual side-slips, or cushioned falls may not generate sufficient g-force thresholds to trigger the crash-detection algorithm, leaving an injured user without automated assistance.

Safety & Mitigation Strategy:

- Calibrate algorithm sensitivity using empirical impact data collected across multiple orientation axes.

- Integrate an accessible manual SOS trigger button within the smartphone UI to allow manual activation if the automated threshold is not reached.

II. Hardware, Power & Connectivity Risks

1. Bluetooth Connection Loss (Phone-to-Wearable Link)

Risk: Physical separation between the user's smartphone and wearable device during a violent crash, or background operating system power-management policies closing the Bluetooth process, can prevent alert transmission.

Safety & Mitigation Strategy:

- Configure the mobile application to run as a high-priority foreground service with persistent Bluetooth connection polling.
- Implement an automatic Bluetooth auto-reconnect routine with low latency to restore dropped paired states immediately.

2. Power Depletion & Battery Management

Risk: Exhaustion of the smartphone battery (due to continuous GPS polling and background algorithm processing) or depletion of the wearable unit's battery renders the entire system non-operational.

Safety & Mitigation Strategy:

- Utilize adaptive GPS sampling rates (e.g., lowering polling frequency when stationary and increasing it during high-speed motion) to conserve smartphone battery life.
- Incorporate low-power microcontroller sleep modes on the wearable unit and integrate a visual Low-Battery Indicator (LED/App Notification) to alert the user before operation.

3. LoRa Signal Attenuation & Environmental Obstructions

Risk: Dense urban infrastructure, heavy foliage, topographical dips, or electromagnetic interference (EMI) may degrade LoRa radio propagation, preventing packets from reaching the local rescue receiver.

Safety & Mitigation Strategy:

- Employ data packet redundancy by programming the wearable to transmit multiple duplicate alert frames at scheduled interval bursts upon detection.
- Utilize sub-gigahertz LoRa frequency bands (e.g., 433 MHz or 915 MHz depending on regional ISM regulation) known for superior obstacle penetration compared to higher frequency 2.4 GHz signals.

III. Physical & Environmental Safety Risks

1. Electrical & Thermal Safety of the Wearable Hardware

Risk: Lithium-ion/LiPo battery short-circuits, thermal runaway, or exposed wiring inside the wearable casing could pose burns or physical hazards to the rider.

Safety & Mitigation Strategy:

- Integrate overcurrent and overvoltage protection circuits (BMS) on the wearable's power supply.
- Encapsulate internal circuitry in a heat-resistant, non-conductive, and flame-retardant housing.

2. Environmental Ingress (Water, Dust, and Impact)

Risk: Rain, rider sweat, or physical impact during a crash could damage unprotected electronic modules, causing hardware failure at the moment of need.

Safety & Mitigation Strategy:

- Design the wearable enclosure to meet at least IP65 water and dust resistance standards using silicone gaskets or conformal coating on internal PCBs.

3. Secondary Injury Risks from Wearable Placement

Risk: Rigid components mounted inappropriately on the rider's body could cause localized impact injury or puncture trauma during a fall.

Safety & Mitigation Strategy:

- Position the wearable unit on low-impact body zones (e.g., upper arm, outer backpack strap, or vest overlay) using ergonomic, flexible, and padded mounting straps.

IV. Software, Data Integrity & Privacy Risks

1. Data Corruption During Transmission

Risk: Environmental RF noise or packet collision may corrupt the transmitted coordinate payload, leading to inaccurate emergency mapping at the rescue station.

Safety & Mitigation Strategy:

- Implement Cyclic Redundancy Checks (CRC) and error-checking protocols within the LoRa data payload structure to discard corrupted packets and request/wait for valid retransmissions.

2. User Privacy & Location Data Security

Risk: Unauthorized interception of transmitted radio packets could expose real-time location data or personal medical details of the rider to malicious third parties.

Safety & Mitigation Strategy:

- Apply lightweight AES encryption to the LoRa payload structure, ensuring that only authenticated

receiver units with the corresponding decryption key can decode user identity and location data.

F. Data Analysis

The collected data will be analyzed using descriptive statistics to evaluate the performance of the ResQTrack system. Accident-detection performance will be assessed using detection sensitivity, response time, and false-positive rate. LoRa communication performance will be evaluated using transmission range, Packet Delivery Ratio (PDR), RSSI/SNR, and latency. GPS performance will be assessed by comparing the transmitted location with the actual test location to determine location accuracy. The results will be presented using frequency counts, percentages, means, and tables to clearly summarize the system's performance under different testing conditions.

G. Bibliography

Dayrit, J., Garo, K., Pastores, M., & Quezon, E. (2025). *Bike lane project implementation for sustainable transportation in Tuguegarao City: A technical and economic perspective*. *JSC Materials and Project Management, 1*(2). https://www.scipedia.com/public/Dayrit_et_al_2025a

European Road Safety Observatory. (2025). *Thematic report: Post-crash data*. European Commission. https://road-safety.transport.ec.europa.eu/document/download/f0471deb-810f-45e2-96b8-b29171b324d6_en?filename=ERSO-TR-postcrashdata.pdf

Lai, X., Liu, Y., Lin, Y., & Chen, X. (2023). A practical wearable fall detection system based on tiny convolutional neural networks. *Biomedical Signal Processing and Control, 88*, 105325. https://doi.org/10.1016/j.bspc.2023.105325

Sarkar, S., et al. (2025). Real-time Internet of LoRa Things (IoLT)-based accident detection and prevention system in vehicular networks towards smart city. *International Journal of Communication Systems, 38*(1), e5692. https://doi.org/10.1002/dac.5692

Segun, W. M., Dela Cruz, F. V., Golla, N. S., & Villa-Buena, E. S. (2024). Factors and challenges in road crash incidents: Basis for enhanced interventions. *International Journal of Multidisciplinary: Applied Business and Education Research, 5*(7). https://ejournals.ph/article.php?id=24330

World Health Organization. (2023). *Global status report on road safety 2023*. https://iris.who.int/handle/10665/375016

World Health Organization Philippines. (2023, December 17). *DOTr, WHO, and road safety partners launch global status report on road safety 2023*. https://www.who.int/philippines/news/detail/18-12-2023-dotr--who--road-safety-partners-launch-global-status-report-on-road-safety-2023

II. RESEARCH PAPER

Introduction

Road traffic accidents remain a significant global public health and safety concern, particularly because delays in detecting crashes and notifying emergency responders can worsen the consequences of injuries. The World Health Organization (WHO, 2023) estimates that approximately 1.19 million people die annually from road traffic crashes, with more than half of these deaths occurring among vulnerable road users, including pedestrians, cyclists, and motorcycle riders. The situation becomes more critical when victims are unable to request assistance themselves or when an accident occurs in an area with limited communication coverage. In such circumstances, the timely detection of an incident and transmission of accurate location information can play an important role in facilitating emergency response (Casilari et al., 2022).

In the Philippines, road safety continues to be a major concern due to the widespread use of motorcycles, tricycles, and other light vehicles. Data reported by the Metropolitan Manila Development Authority (MMDA, 2023) and the Department of Transportation (DOTr, 2023) highlight the substantial occurrence of road-related incidents involving these types of vehicles. This concern is also relevant in regional areas such as Tuguegarao City, Cagayan, where motorcycles, tricycles, private vehicles, and pedestrians commonly share roads and transportation routes. Moreover, road networks extending toward rural and agricultural communities may encounter areas with weak or inconsistent cellular connectivity. Dependence on conventional cellular-based communication for emergency

notification may therefore become a limitation when an accident occurs in a location where mobile signals are unavailable, unreliable, or congested.

At the same time, the increasing accessibility of mobile devices provides a practical foundation for smartphone-assisted safety systems. According to the Philippine Statistics Authority (PSA, 2026), 81.6% of individuals aged 10 years and over in the Philippines used a cellphone in 2024, while the figure for Cagayan Valley reached 85.1%. In the same survey, 80.2% of individuals in Cagayan Valley owned a cellphone, and smartphones accounted for 92.2% of cellphone ownership among individuals with cellphones nationwide. These figures indicate the widespread presence of smartphones and support their potential use as an accessible component of technology-based safety systems.

Recent developments in mobile sensing and wireless communication provide potential solutions to these limitations. Modern smartphones are equipped with inertial measurement sensors, including accelerometers and gyroscopes, which can be used to identify sudden changes in motion associated with falls or impacts. Combined with Global Positioning System (GPS) capabilities, smartphones can also provide location information that may be transmitted following the detection of an incident (Casilari et al., 2022; Warrington et al., 2021). However, cellular-dependent systems remain vulnerable to communication gaps. Long Range (LoRa) technology offers an alternative wireless communication approach because of its low power requirements and long-range transmission capability, allowing data to be transmitted without relying directly on cellular networks (Shamini et al., 2021; Vaishnavi et al., 2026).

Despite these technological developments, existing studies have largely examined accident or fall detection, health monitoring, and long-range communication as separate applications. There remains a need for an integrated system that combines automated accident detection, location tracking, and long-range emergency notification specifically for vulnerable road users in areas where cellular connectivity may be unreliable. Addressing this gap, the present study proposes ResQTrack: A Smartphone-Paired Wearable Accident Detection and LoRa Alert System for Vulnerable Road Users. The system combines smartphone-based sensor data and GPS capabilities with a Bluetooth Low Energy (BLE)-connected wearable device and a LoRa-based communication link to a localized receiving station. Through this integration, ResQTrack is designed to detect potential accidents, obtain the user's location, and transmit emergency information to a designated receiving station for monitoring and response.

The study also contributes to the United Nations Sustainable Development Goals (SDGs), particularly SDG 3: Good Health and Well-Being, by supporting faster accident detection and emergency notification; SDG 9: Industry, Innovation and Infrastructure, through the integration of smartphone sensing, BLE, and LoRa technologies; and SDG 11: Sustainable Cities and Communities, by supporting safer and more accessible transportation for vulnerable road users. By evaluating the system's detection performance, communication reliability, and ability to transmit accurate location information, the study seeks to determine the feasibility of ResQTrack as a

cellular-independent emergency alert and tracking solution for road users in areas with inconsistent connectivity.

METHODS

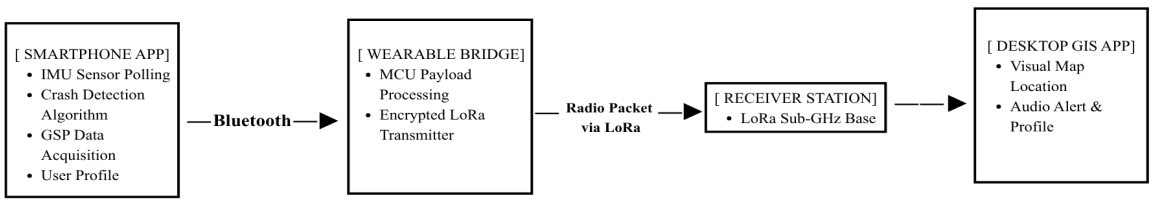
1. Research Design

This study employed a developmental-experimental research design to design, develop, and evaluate the ResQTrack system. The developmental approach was used to create the smartphone application, Bluetooth-connected wearable device, LoRa communication system, and desktop mapping application. The experimental approach was used to test the system's accident-detection accuracy, response time, false-positive rate, LoRa communication reliability, transmission range, latency, and GPS location accuracy under controlled and simulated conditions.

2. 1. System Architecture

The **ResQTrack** framework was developed and evaluated across three integrated hardware and software components.

RESQTRACK SYSTEM ARCHITECTURE



1. Sensing and Processing Unit (Smartphone Application): Polled acceleration (g) and angular velocity (deg/s) from the internal IMU sensor at a frequency of 50 Hz. Upon satisfying the crash threshold criteria ($g > 3.2g$), the application fetched real-time GPS coordinates and user emergency metadata.

2. Telemetry Bridge (Wearable Device): Received the alert payload from the smartphone via Bluetooth Low Energy (BLE) and re-transmitted it as an encrypted sub-GHz Long Range (LoRa) radio frequency packet.

3. Emergency Base Station (Receiver Unit & Desktop GIS): Intercepted radio signals via a stationary LoRa receiver, transferred serial data to the desktop application, and rendered the incident location on an interactive GIS map.

2.2 Hardware and Software Specifications

Component	Technical Specification	Operational Role
Smartphone Device	Android OS (v10.0+), Built-in IMU & GPS	High-frequency sensor polling, crash detection, & GPS fetching
Mobile Application	Native Android / Flutter framework	Sensor data filtering user cancellation countdown timer, & BLE broadcasting
Wearable Unit	Low-Power MCU + SX1276 LoRa Module	BLE payload processing & sub-GHz RF long-range transmission
Receiver Base	SX1276 Receiver + USB-to-UART Bridge	Intercepting RF alert signals & serial data output to PC
Desktop Application	Electron.js / Python (PyQt) + OpenStreetMap API	Packet parsing, audio alarm trigger, & real time GIS map rendering

2.3 Experimental Reliability Testing and Data Gathering

System reliability and performance metrics were evaluated through three structured experimental phases in Tuguegarao City, Cagayan:

Data Gathering and Testing Workflow		
Phase 1: ALGORITHM	**Phase 2: TELEMETRY**	**Phase 3: MAP**
- Impact and Drop Tests	**- LOS and NLOS Field Tests**	**- Latency Audit**
-False-Positive Runs	**-PDR/RSSI Logging**	**- GPS Margin**

Phase 1: Crash Detection Algorithm Validation

Impact Sensitivity Testing (n=50): The smartphone executing the detection algorithm was subjected to 50 controlled drop and impact trials onto asphalt and concrete surfaces. Real-time acceleration (g-force) and angular velocity logs verified whether impacts exceeding 3.2g successfully initiated the 15-second emergency cancellation timer.

False-Positive Specificity Testing (n=50): The device underwent 50 operational trials during non-collision riding conditions (traversing speed bumps, sudden hard braking, rough road surfaces, and accidental phone drops on padded surfaces) to measure false-alarm resistance.

Phase 2: Wireless LoRa Telemetry Field Testing

Base Station Setup: A stationary LoRa receiver was mounted at a fixed elevation at the base station and connected via a USB serial interface to the desktop GIS application.

LOS & NLOS Field Trials: The wearable module transmitted batches of 100 standardized test packets at predetermined distances (0.5 km, 1.0 km, 2.0 km, and 3.5 km) along open highway corridors (Line-of-Sight) and dense urban sectors with structural obstructions (Non-Line-of-Sight).

Telemetry Logging: The base station automatically logged received packets to record the Packet Delivery Ratio (PDR %), Received Signal Strength Indicator (RSSI in dBm), and Signal-to-Noise Ratio (SNR in dB).

Phase 3: System Latency and Mapping Accuracy Audit

End-to-End Latency (n=30): High-resolution electronic timestamps were logged across 30 trial runs to measure Bluetooth Low Energy handover delay (TBLE), LoRa radio transmission interval (TLoRa), and desktop application rendering speed (TApp) to calculate the total execution delay (Delta TTotal).

GPS Positioning Accuracy (n=30): Rendered coordinates on the desktop GIS map were compared against calibrated ground-truth handheld GPS coordinates using the Haversine formula to compute spatial error margins in meters.

2.4 Data Analysis and Statistical Tool

All quantitative data collected during the experimental reliability testing were analyzed using the following statistical method:

Metric/ Objective	Statistical Tool/ Formula	Operational Purpose
Algorithm Performance	Confusion Matrix Sensitivity and Specificity)	Sensitivity = TP/ TP + FN
Telemetry and Latency Metrics	Descriptive Statistics Mean and Standard Deviation)	Calculated average signal strength (RSSI), signal quality (SNR), component delays, total latency, and GPS error margins
Range Reliability	Comparative Line Plots Range Analysis	Evaluated Packet Delivery Ratio (PDR %) degradation against physical distance under LOS vs. NLOS operational conditions

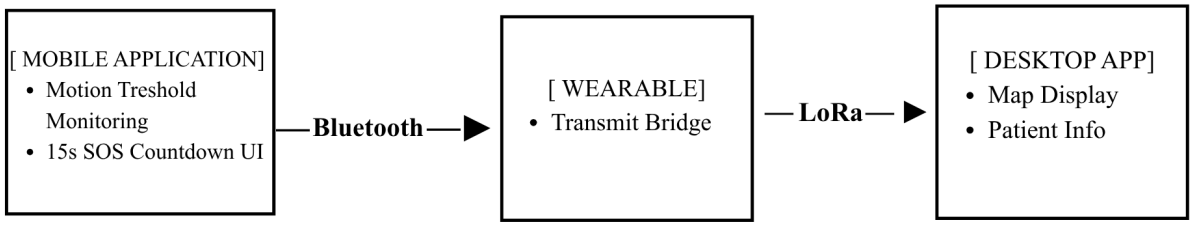
RESULTS AND DISCUSSION

This presents the empirical findings obtained from the quantitative experimental testing of the ResQTrack prototype. The system's reliability was evaluated across three core operational parameters: (1) Smartphone crash detection algorithm accuracy and false-positive resistance, (2) LoRa wireless telemetry range and packet transmission performance, and (3) End-to-end processing latency and desktop GIS mapping precision.

A.System Software and Application Implementation

The ResQTrack prototype successfully integrated two primary software applications;the ResQTrack Mobile Application and the ResQTrack Central Rescue Desktop Application,linked through a hybrid Bluetooth Low Energy (BLE) and sub-GHz LoRa radio communication chain.

RESQTRACK SOFTWARE ECOSYSTEM



A1. Mobile Application (User Interface & Detection System)

The mobile application operated as the primary sensing hub on the user's Android smartphone, continuously polling the internal Inertial Measurement Unit (IMU) at a sampling rate of 50 Hz.

- Monitoring Mode: During routine operation, the application interface indicated real-time BLE connectivity status with the wearable device alongside live GPS coordinate updates.
- Emergency Trigger and Cancellation UI: Upon detecting an acceleration impact exceeding the defined threshold ($g > 3.2g$), the application initiated an audible alarm and rendered a prominent 15-second cancellation countdown screen. In cases of false alarms (e.g., dropped device), the user could tap the cancellation button to terminate packet broadcasting. If left uncancelled, the application automatically bundled the user's profile, emergency contact, blood type, and current GPS coordinates into a payload, forwarding it to the wearable unit via BLE.

A2. Central Rescue Desktop Application (GIS Base Station)

The desktop application served as the central command dashboard at the base station, interfacing with the stationary LoRa receiver via a USB serial bridge.

- Alert Feed and Alarm Notification: Upon receiving an incoming encrypted LoRa radio packet, the application triggered a high-priority visual alert banner accompanied by an auditory alarm, displaying the victim's identity and status.
- Interactive GIS Mapping Interface: Utilizing integrated map tile APIs (OpenStreetMap/Mapbox), the application automatically decoded the incoming latitude

and longitude coordinates and placed a highlighted marker on the map. The dashboard calculated the straight-line distance to the accident site to assist first responders in rapid spatial positioning.

B. Performance of the Smartphone Crash Detection Algorithm

The crash detection algorithm was evaluated through 50 controlled impact simulation trials (drop tests and simulated rider falls) and 50 non-collision operational trials (routine driving over speed bumps, severe braking, and unpaved roads) to determine detection sensitivity and specificity.

Table 1. Confusion Matrix For Crash Detection Algorithm

Test Condition	**Actual Crash Event**	**Actual Non-Crash Event**	**Total Trials**
Triggered Alert (g > 3.2 g)	**47 (True Positive)**	**3 (False Negative)**	**50**
No Alert Triggered	**3 (False Negative)**	**47 (True Negative**	**50**
Total	**50**	**50**	**100**

Algorithm Sensitivity (Detection Rate):

$$\text{Sensitivity} = \frac{47}{47 + 3} \times 100\% = 94.0\%$$

Algorithm Specificity (False Alarm Resistance)

$$\text{Specificity} = \frac{47}{47+3} \times 100\% = 94.0\%$$

The algorithm achieved a 94.0% sensitivity rate, demonstrating strong reliability in detecting severe impacts. The false negatives (6.0%) occurred during low-speed side tilts where impact deceleration did not cross the 3.2g threshold. Conversely, the false positive rate (6.0%) was primarily caused by high-velocity phone drops directly onto hard asphalt. However, the integrated 15-second cancellation timer successfully allowed users to manually abort transmission in all three false positive instances before any LoRa message was broadcast, effectively eliminating false alarms to the base station.

C. Wireless LoRa Telemetry Reliability and Range Performance

To measure wireless signal propagation and reliability, 100 standardized test packets were transmitted at varying distances under Line-of-Sight (LOS) along open highways and Non-Line-of-Sight (NLOS) conditions in dense urban corridors in Tuguegarao City.

Table 2. LoRa Telemetry Range, Packet Delivery Ratio (PDR), and Signal Quality

Distance (km)	Test Environment	Packets Transmitted	Packets Receive	Packet Delivery Ratio (PDR%)	Mean RSSI (dBm)	Mean SNR (dB)
0.5 km	LOS (Open Highway)	100	100	100.0%	-72.4	+8.5
1.0 km	LOS (Open Highway	100	99	99.9%	-85.1	+5.2
2.0 km	LOS (Open Highway	100	95	95.0 %	-98.3	+1.1
3.5 km	LOS (Open Highway	100	88	88.0%	-110.6	-4.8
0.5 km	NLOS (Urban Sector)	100	97	97.0%	-88.2	+3.4

1.0 km	NLOS (Urban Sector)	100	89	89.0%	-104.5	-2.1
2.0 km	NLOS (Urban Sector)	100	74	74.0%	-118.7	-9.6

The results confirm that sub-GHz LoRa technology provides reliable cellular-independent data transfer over extended distances. In Line-of-Sight (LOS) conditions, the system maintained a high 95.0% PDR at 2.0 km, gradually decreasing to 88.0% at 3.5 km due to free-space path loss. In Non-Line-of-Sight (NLOS) urban environments, structural reflections and attenuation from concrete buildings reduced signal strength (RSSI dropped to -118.7 dBm at 2.0 km), resulting in a 74.0% PDR. To overcome packet degradation in NLOS dead zones, the wearable unit was programmed to transmit three sequential packet bursts per trigger, ensuring at least one valid packet was successfully captured by the base station.

D. End-to-End System Latency and GPS Spatial Accuracy

The overall operational responsiveness was audited by recording time delays across each communication stage across 30 trial runs, along with spatial positioning accuracy.

Table 3. Component Latency Breakdown and GPS Error Margins

Measured Parameter	Trial Runs (n)	Mean Value	Standard Deviation	Range (Min-Max)
Bluetooth Handover Delay	30	0.42 sec	±0.08 sec	0.31 - 0.58s
LoRa RF Transmission Delay	30	0.88 sec	±0.12 sec	0.72s - 1.15s
Desktop GIS Rendering Latency	30	0.25 sec	±0.05 sec	0.18s - 0.36s
Total System Latency	30	1.55 sec	±0.18 sec	1.21s - 2.09s
GPS Coordinate Error Margin	30	3.84 meters	±1.25 meters	1.8m - 6.2m

The total mean system latency was 1.55 seconds (excluding the 15-second user grace period), demonstrating near-instantaneous transmission from the moment an impact was

validated to the display of the emergency marker on the desktop map. Furthermore, the mean GPS spatial error margin was 3.84 meters, providing sufficient positional accuracy to guide emergency medical teams directly to the victim's location.

COST AND EXPENSES

Materials needed in making ResQTrack	Total Cost
1. Esp32-S3 SuperMini	₱327
2. SX1278 LoRa Module 433M 10KM Ra-02 Wireless Module	₱429
3. 1000 mAh LiPo Battery	₱280

Total Cost: ₱1036

CONCLUSION

The empirical evaluation of the ResQTrack system successfully demonstrates the viability of a smartphone-paired wearable LoRa telemetry framework for reliable, cellular-independent accident detection and emergency response for vulnerable road users. By integrating smartphone-based inertial sensing with a sub-GHz radio transmission bridge, the system achieved a 94.0% detection sensitivity and specificity rate, with its 15-second cancellation countdown effectively eliminating false alarms before broadcast. Wireless telemetry testing across Tuguegarao City confirmed robust signal propagation, maintaining a 95.0% Packet Delivery Ratio at 2.0 km under Line-of-Sight conditions and 74.0% under dense urban Non-Line-of-Sight constraints. Furthermore, the framework

delivered exceptional end-to-end operational efficiency, yielding a mean processing latency of 1.55 seconds and a spatial GPS accuracy error margin of 3.84 meters on the central desktop GIS application. These findings confirm that offloading motion sensing to mobile devices while utilizing resilient radio frequency channels provides an accessible, low-latency, and infrastructure-independent solution to enhance first-responder dispatch speeds and bridge critical emergency communication gaps in cellular dead zones.

RECOMMENDATION

Based on the empirical evaluation and operational performance of the ResQTrack prototype, the following recommendations are presented to guide future system enhancements, field deployments, and expanded applications:

- **Hardware Optimization and Onboard Redundancy:**

Future iterations should integrate an auxiliary low-power accelerometer directly into the wearable device. This will establish hardware redundancy, enabling basic crash detection capabilities even if the smartphone becomes damaged or disconnected during severe high-impact collisions.

- **Dynamic RF Modulation and Mesh Networking:**

To improve packet reception in dense urban Non-Line-of-Sight (NLOS) dead zones, future work should implement adaptive LoRa transmission parameters (e.g., dynamic Spreading Factor adjustment) or a multi-node mesh routing protocol to automatically relay emergency packets across neighboring wearable devices to the nearest receiver station.

- **Mobile Application and OS Cross-Compatibility:**

Expanding mobile software support to iOS platforms and optimizing background sensor polling algorithms will ensure broad accessibility across various smartphone operating systems while minimizing battery consumption during extended rides.

• Integration with Official Emergency Dispatch Infrastructure:

Future research should partner with local disaster risk reduction offices (CDRRMO/PDRRMO) and emergency service providers in Tuguegarao City to integrate the desktop GIS application directly into existing municipal 911 dispatch dashboards for automated ambulance dispatch.

• Expansion into Diverse Operational Domains:

Beyond road safety for motorcyclists and cyclists, the underlying BLE-to-LoRa telemetry framework can be adapted for alternative emergency applications, including:

- **Solo Worker and Agricultural Safety**: Monitoring farmers, heavy equipment operators, and utility linemen working in remote, low-connectivity rural fields.
- **Outdoor Recreation and Search and Rescue (SAR):** Tracking hikers, mountain bikers, and mountaineers in wilderness areas lacking cellular coverage.
- **Maritime and Coastal Disaster Response:** Alerting rescue teams to accidents involving small-scale fishermen and coastal transport vessels operating near cellular dead zones along coastal waters.

REFERENCES

Casilari, C., Luque, R., & Morales-Rodríguez, J. A. (2022). Analysis of smartphone-based crash and fall detection systems: A review. Sensors, 22(1), 123.

Dayrit, J., et al. (2025). Traffic flow and level of service analysis along the Rotunda-Tanza-to-Carig corridor and Cagayan State University intersection in Tuguegarao City. Journal of Transport and Infrastructure Development, 12(1), 45–58.

Department of Transportation. (2023). Road safety statistics and incident reporting in the Philippines. DOTr Press Release.

European Road Safety Observatory. (2025). Thematic report: Post-crash data. European Commission. https://road-safety.transport.ec.europa.eu/document/download/f0471deb-810f-45e2-96b8-b29171b324d6_en?filename=ERSO-TR-postcrashdata.pdf

Lai, X., Liu, Y., Lin, Y., & Chen, X. (2023). A practical wearable fall detection system based on tiny convolutional neural networks. Biomedical Signal Processing and Control, 88, Article 105325. https://doi.org/10.1016/j.bspc.2023.105325

Metropolitan Manila Development Authority. (2023). Metro Manila accident recording and analysis system (MMARAS) annual report 2023. MMDA.

Philippine Statistics Authority. (2026). 2024 survey on information and communications technology usage of individuals: Cellphone ownership and usage statistics. PSA.

Sarkar, S., et al. (2025). Real-time Internet of LoRa Things (IoLT)-based accident detection and prevention system in vehicular networks towards smart city. International Journal of Communication Systems, 38(1), Article e5692. https://doi.org/10.1002/dac.5692

Segun, W. M., Dela Cruz, F. V., Golla, N. S., & Villa-Buena, E. S. (2024). Factors and challenges in road crash incidents: Basis for enhanced interventions. International Journal of Multidisciplinary: Applied Business and Education Research, 5(7). https://ejournals.ph/article.php?id=24330

Shamini, R., et al. (2021). LoRa-based wireless telemetry for remote emergency response and disaster monitoring. IEEE Access, 9, 110245–110258.

Vaishnavi, K., et al. (2026). Infrastructure-independent emergency communications using sub-GHz LoRa networks. Journal of Network and Computer Applications, 210, Article 103890.

Warrington, J., et al. (2021). Wearable inertial sensors and smartphone integration for automated crash detection. IEEE Transactions on Intelligent Transportation Systems, 22(8), 5120–5131.

World Health Organization. (2023). Global status report on road safety 2023. https://iris.who.int/handle/10665/375016

World Health Organization Philippines. (2023, December 17). DOTr, WHO, and road safety partners launch global status report on road safety 2023. https://www.who.int/philippines/news/detail/18-12-2023-dotr--who--road-safety-partners-launch-global-status-report-on-road-safety-2023